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SURVEY

Emerging SMOTE and GAN Variants for Data Augmentation in Imbalance Machine Learning Tasks: A Review

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ABSTRACT Class imbalance is a pervasive challenge in real-world machine learning (ML) applications, where the minority class, often the class of interest, is significantly underrepresented. This imbalance can degrade model performance, result in misleading evaluation metrics, and complicate validation processes. Two prominent data-augmentation techniques to address class imbalance are the Synthetic Minority Oversampling Technique (SMOTE) and Generative Adversarial Networks (GAN). However, both techniques have inherent limitations, motivating the emergence of novel variants designed to overcome these challenges. While previous reviews have typically focused on specific domains, conventional methodologies, or broad strategy overviews, this review presents a unified taxonomy that outlines the causes, types, and implications of class imbalance across diverse ML tasks. It further examines emerging trends in the application of SMOTE and GAN techniques, their limitations, and hybrid adaptations. By categorising imbalance types and analysing models, metrics, datasets, and comparative approaches, this review provides actionable insights and identifies future research directions for practitioners and researchers working to address class imbalance in real-world ML tasks.

INDEX TERMS Class imbalance, data-augmentation, generative adversarial networks, machine learning, SMOTE.

I. INTRODUCTION

A dataset with a skewed distribution between majority and minority samples is often described as exhibiting class imbalance. This issue is prevalent in many real-world scenarios, such as in fraud detection [1], [2], where fraudulent transactions are much rarer than legitimate ones; medical diagnosis [3], [4], [5], [6], where some diseases occur infrequently; aero-engine fault prediction [7], [8], [9], [10]; and industrial material forecasting [11], [12], [13], [14], [15]. Class imbalance poses significant challenges during

model training, often biasing the model towards the majority class. This reduces the model's ability to accurately predict instances of the minority class, ultimately diminishing the overall performance of machine learning (ML) models.

Addressing class imbalance is critical because key factors that impact model performance, evaluation metrics, and validation process. ML models are typically optimised for balanced datasets, making them inherently biased towards the majority class [16], [17]. This bias can lead to poor performance in recognising patterns in the minority class. For example, in medical diagnosis, a model might accurately identify common conditions but struggle to detect rare diseases, resulting in missed diagnoses or false negatives.

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Similarly, in fraud detection, the model may frequently overlook fraudulent transactions because the dataset is dominated by legitimate ones, thereby increasing the risk of undetected fraud.

Standard performance metrics such as accuracy can be misleading in the presence of class imbalance. A model might achieve high overall accuracy simply by consistently predicting the majority class, while failing to identify instances from the minority class. For instance, in a credit scoring system where defaults are rare, a model might report high accuracy yet fail to predict actual defaults, making it ineffective for risk management. This example illustrates how relying solely on accuracy can obscure the true performance of a model in critical areas. Validating models in the presence of class imbalance presents significant challenges, especially where there exists a scarcity in minority class samples in the validation set. In wildlife monitoring, for example, models may appear to perform well simply because the dataset is dominated by common species, while rare, endangered species remain underrepresented and poorly predicted. This can cause the model to rely on the majority class (non-endangered species) for its predictions, inflating the perception of its ability to generalise to unseen data. Even metrics designed for imbalanced datasets may fail to fully address this issue, as they might not effectively capture the model's effectiveness on minority classes (endangered species) [18], [19].

Multiple strategies have been proposed for addressing class imbalance in ML problems and these can be divided into two primary categories, namely algorithm-level and data-level methods. Algorithm-level techniques [20], [21] focus on modifying how the model learns from the dataset and evaluation metrics in tackling issues that influence model performance due to imbalance. Techniques under algorithm-level include cost-sensitive learning [22], [23], [24], [25], [26] boundary shifting, multi-objective optimisations, ensemble and kernel specific methods. Algorithm-level techniques however risk overfitting the minority class and faces challenges in accurately quantifying misclassification costs, while ensemble methods can become computationally expensive and still struggle with extreme imbalances [27], [28]. These limitations highlight the need for data-augmentation techniques, and hybrid approaches that leverage the strengths of algorithm-level techniques in generating synthetic samples to balance datasets, improving model generalisation, and reducing overfitting, thus providing a more effective solution for handling class imbalance in real-world ML tasks [29], [30]. Two emerging techniques that have received significant focus for addressing class imbalance in ML tasks are Synthetic Minority Oversampling Technique (SMOTE) and Generative Adversarial Networks (GAN). This review not only considers the characteristics and applications of SMOTE and GAN but also delves into their limitations, hybrid variants, and emerging trends, offering fresh insights into their potential for addressing class imbalance in ML tasks.

While previous reviews have significantly advanced the understanding of class imbalance, many have understandably concentrated on specific application domains, such as medicine or fault diagnosis, or have focused on particular strategies like cost-sensitive learning [22], [31], [32]. Broader surveys and more recent reviews [33], [34], [35], [36] have provided valuable overviews of current approaches. However, these often offer limited coverage of emerging data-augmentation techniques—such as SMOTE and GAN variants—and do not fully consolidate foundational insights into the causes, types, and implications of class imbalance. This review fills that gap by presenting a harmonised taxonomy that captures the underlying characteristics of imbalance—its origins, forms, and effects—across diverse ML contexts. To the best of our knowledge, this is the first review to systematically establish these concepts alongside an in-depth exploration of emerging SMOTE and GAN variants. This approach not only clarifies key foundational issues but also lays the groundwork for more informed applications of data-augmentation strategies in real-world ML tasks.

The rest of this work is structured as follows: Section II highlights the strategy and research questions that guides the review, while Section III presents a taxonomy of class imbalance, outlining its causes, types, and implications. Data-augmentation techniques are discussed in Section IV and performance metrics for evaluating models for the emerging variants are presented in Section V. Discussion and future studies are provided in Sections VI, with conclusions in Section VII.

II. REVIEW STRATEGY

This review explores the growing field of data augmentation for class imbalance in ML, with a specific focus on emerging SMOTE and GAN variants. To ensure a comprehensive and current analysis, studies were sourced from leading academic databases, including IEEE Xplore, SpringerLink, Elsevier (ScienceDirect), MDPI, Scopus, and Google Scholar. The review emphasises recent advancements, prioritising works published between 2020 and 2024 to capture the latest innovations and insights. The search process employed targeted keywords such as “class imbalance,” “imbalanced data,” “data augmentation,” “SMOTE,” “GAN,” “synthetic sample generation,” “oversampling,” “undersampling,” “hybrid techniques,” and “imbalanced classification.” A total of 61 peer-reviewed studies were selected, many of which are indexed in high-impact journals listed in Scimago Journal Rank (ScimagoJR), ensuring both quality and relevance. These references encompass foundational methods, recent adaptations, and domain-specific innovations, offering a well-rounded and in-depth overview of the field. This strategy not only presents a clear picture of the current state of data augmentation in imbalanced learning tasks but also highlights existing research gaps, setting the stage for future exploration.

This study addresses key research questions (RQ) that encompass critical aspects of the imbalanced data problem

and the role of emerging SMOTE and GAN variants in data augmentation:

- 1) **RQ1:** What are the causes, types, and implications of class imbalance in ML?
- 2) **RQ2:** What are the emerging trends in SMOTE and GAN-based data augmentation for addressing class imbalance in real-world ML tasks?
- 3) **RQ3:** How do SMOTE and GAN-based variants compare in terms of effectiveness, adaptability, and performance when applied to imbalanced ML tasks across different domains?
- 4) **RQ4:** How is the performance of emerging data-augmentation variants evaluated, and what practical limitations and opportunities for future research have been identified?

Section III of this review answers the RQ1, while Section IV offers insights into RQ2 and RQ3. Finally, Sections V and VI of the review addresses RQ4 by highlighting how performance is measured and outlining potential areas for continued research.

III. TAXONOMY OF IMBALANCE: CAUSES, TYPES, AND IMPLICATIONS

Class imbalance in real-world scenarios arises from various factors tied to the characteristics of specific application domains. The rarity of certain events or conditions—such as diseases, manufacturing defects, or fraudulent transactions—naturally leads to fewer instances of the minority class [1], [2], [3]. In the medical domain, for example, datasets often overrepresent prevalent conditions while underrepresenting rarer but clinically significant ones [3], [5]. Similarly, in fraud detection, the vast majority of transactions are legitimate, resulting in highly skewed datasets [1], [2]. Data collection practices can also introduce or exacerbate imbalance. These include oversampling of majority classes or under-sampling of minority classes—either intentionally for efficiency or inadvertently during preprocessing. In high-risk domains such as aero-engine fault diagnosis, collecting real-world fault data is particularly resource-intensive, which limits the availability of minority class samples [7], [9], [10].

In some cases, events like seasonal illnesses or context-specific fraud occur only under particular conditions, resulting in temporary or situational imbalances if the data collection is not comprehensive. Other domains, such as customer segmentation or object detection, may exhibit intrinsic or evolving imbalance. For instance, a small subset of customers may account for most purchases, or some object categories may occur much less frequently in visual datasets [16]. Recognising these domain-specific causes of class imbalance is critical for selecting appropriate data-augmentation strategies, ensuring robust and generalisable ML models.

From a typological standpoint, one form of imbalance is known as *Intrinsic imbalance*, which stems from the natural occurrence of the dataset being studied. This is

especially common in medical datasets, where the number of healthy patients far exceeds those diagnosed with a specific condition. Also, certain disease condition being studied (e.g. Huntington's disease) are inherently rare when compared to more common conditions like hypertension [37], [38]. In such cases, the imbalance in the dataset reflects the actual distribution of conditions in the real-world. By contrast, *Extrinsic imbalance* is introduced by external factors related to data collection or preprocessing methods [39], [40]. For instance, an online survey may disproportionately attract younger participants who are more familiar with digital platforms, thereby underrepresenting older individuals. In this scenario, the imbalance is introduced by the data collection method rather than the underlying data distribution.

TABLE 1. Sample distribution for global imbalance.

Classes	Instances	Engine condition
N	1,000	Normal operating condition
A	500	Fuel system fault
B	80	Cooling system fault
C	20	Turbine blade damage

Two other notable categories are binary imbalance and multi-class imbalance. *Binary imbalance* occurs in datasets with only two classes, where one class significantly outnumbers the other. A typical example of binary imbalance is spam detection, where spam emails (the minority class) occur far less frequently than legitimate emails (the majority class) [41]. In contrast, *Multi-class imbalance* arises in datasets with more than two class distribution, where one or more classes are significantly underrepresented. For example, in a facial recognition system trained to identify individuals from different ethnic groups [42], [43], [44], certain ethnic groups may be underrepresented in the training data. This imbalance can result in biased models that perform poorly for those groups [45], [46]. A common approach to handling multi-class imbalance is to convert the problem into a series of binary tasks using a one-vs-all strategy [47], [48], [49]. In this method, samples from a single class are compared against samples from all the other classes combined. However, studies have shown that this strategy is not easily applicable to multi-class imbalance scenarios due to certain unique characteristics of multi-class distributions. For instance, a class may be considered a minority when compared to one class but a majority when compared to another class with comparatively fewer samples. Additionally, there might exist severe overlap in the boundaries of class samples, making the baseline one-vs-all strategies ineffective. In such cases, and with a severe rarity of class samples, applying data-augmentation techniques may be either prove ineffective or even degrade model performance [50], [51], [52], [53].

In multi-class problems, class imbalance can be categorised as global or local. *Global imbalance* refers to disparities that exists across the entire dataset, while *Local imbalance* occurs only under certain data conditions or

TABLE 2. Sample distribution for local imbalance.

Region	Operating condition	Classes: instances
1	Low load operation	N: 600 A: 150
2	High temperature operation	N: 300 A: 200 B: 50 C: 10
3	High altitude operation	N: 100 A: 20 B: 30 C: 10

subsets [19], [20]. Consider a synthetically generated dataset of engine fault detection in TABLE 1 as an illustration. Globally, Class N is the dominant class, vastly outnumbering Classes A, B, and C. Among the fault classes, Class A is most common, while Class C is extremely rare. Global imbalance here means that the model is likely to become biased towards predicting Class N, potentially leading to higher rate of missed detections for the rare but critical faults in Class C. To address global imbalance, techniques such as oversampling Classes A, B, and C or undersampling Class N can be employed. These strategies help the model pay appropriate attention to the rarer but important classes, such as Class C.

Local imbalance can be illustrated by considering a synthetically generated aero-engine fault detection dataset with different operating regions/conditions, as shown in TABLE 2. In Region 2, Class N dominates with 300 instances making the 10 instances in Class C almost negligible. Although Class C is already globally underrepresented, it is also locally scarce in Region 2. A visual depiction of this imbalance is shown in Fig. 1. In Region 3, Class B appears more frequently than Class C, but Class N remains the dominant class. Again, the model may struggle to correctly identify faults in Class C, as the very low number of samples contributes to local imbalance [54], [55]. Locally targeting oversampling in Regions 2 and 3, where Class C is severely underrepresented, can enable the model more effectively recognise faults under specific operating conditions.

Class imbalance is also characterised as absolute and relative. *Relative imbalance* occurs when the minority class instances are significantly fewer to the other classes, often expressed as a low ratio of m/N , where m is the number of minority class samples and N is the total number of validation samples. Conversely, *Absolute imbalance* refers to the scarcity of minority class instances in an absolute sense, where m is intrinsically low in value, regardless of the dataset size [18], [19], [22].

In *Static imbalance*, the degree of imbalance remains constant over time. This can be seen in medical imaging dataset, where the proportion of healthy to diseased samples stays consistent throughout the data collection period. In contrast, *Dynamic imbalance* refers to datasets, where the imbalance changes over time or across different contexts. For instance,

in social media streaming data, certain topics may become dominant only during specific events or periods, leading to time-dependent fluctuations in class distributions [56], [57].

IV. DATA-AUGMENTATION TECHNIQUES

A. COMMONLY USED TECHNIQUES

A common approach of handling class-imbalance problems is *undersampling*, which reduces the number of majority class samples to match the size of the minority class by randomly selecting a subset. However, this method can lead to a loss of valuable information and is only practical when dealing with very large datasets where such a loss is acceptable. An alternative technique is *oversampling*, in which elements of the minority class are duplicated until they match the majority class. This method involves directly replicating samples and does not reflect model generalisability. It is also known to be prone to overfitting [36]. Another common method divides the majority class into three subsets and trains three separate models, each paired with the full minority class. The final prediction is then obtained through a majority vote among the three models. This *ensemble-based approach* aims to reduce the impact of imbalance while preserving more of the original data [58].

B. SMOTE

SMOTE is one of the most widely used solutions in addressing class-imbalance. New synthetic samples of the minority class are constructed by interpolating minority classes in the same neighbourhood [59]. To achieve this, SMOTE randomly selects a minority class sample (x_i) from the minority class X and determines the nearest minority class sample (x'_i) in the surrounding neighbourhood: where $x'_i \in X$. Thereafter, a line is computed between the x_i and x'_i , where a new minority class instance x_{new} is created using the formula:

$$x_{\text{new}} = x_i + (x'_i - x_i) \cdot \delta, \quad \delta \in [0, 1] \quad (1)$$

where δ is a random number between 0 and 1. An illustration of SMOTE is shown in Fig. 2.

A potential drawback of SMOTE however is that the minority instances are generated without reference to the majority class. Another challenge is the introduction of noise by generating data that does not exist in the domain [59], [60], [61], [62]. This necessitates variants of SMOTE that addresses this challenge. György Kovács [63] developed an open-source Python package featuring 85 SMOTE variants and 61 compatible multi-class oversampling techniques. The package includes cross-validation and evaluation tools, enhancing data preprocessing for imbalanced datasets. It enables efficient benchmarking by comparing emerging data-augmentation methods with established techniques, improving robustness and flexibility in ML workflows. Some well-established SMOTE variants included in the package are SMOTE-Edited Nearest Neighbour (ENN), Borderline-SMOTE (B-SMOTE), Support Vector Machine (SVM)-SMOTE and Adaptive Synthetic sampling (ADASYN) [64], [65], [66], [67]. Emerging data-augmentation techniques

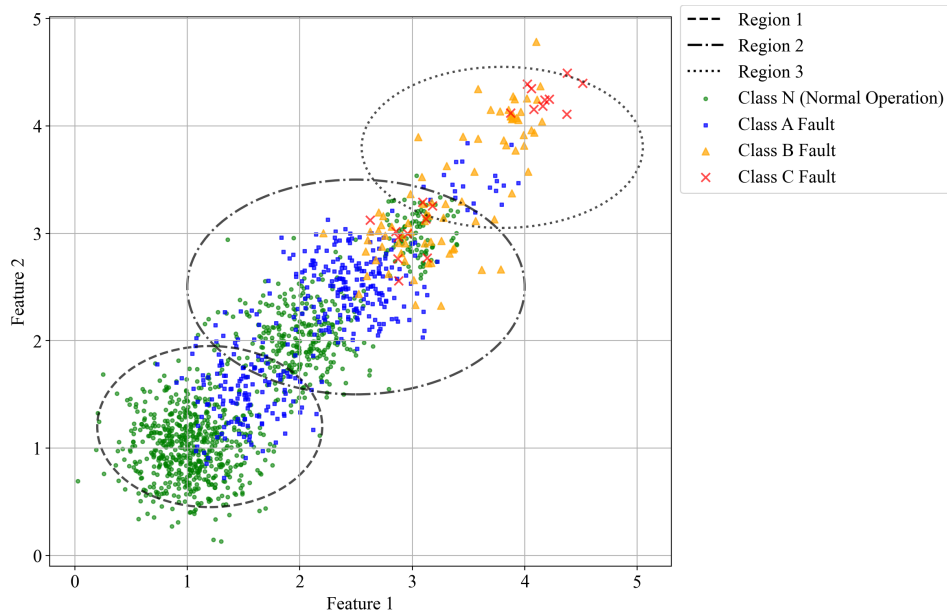


FIGURE 1. Illustration of local class imbalance on aero-engine fault detection dataset.

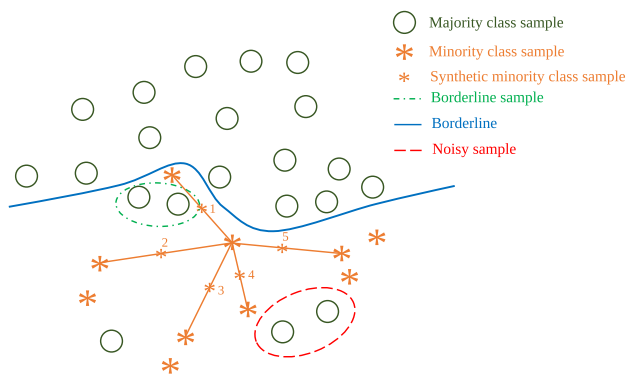


FIGURE 2. An illustration of data-augmentation using SMOTE.

have mostly featured variants of SMOTE [68], [69] and GAN [70], [71], [72], [73] as the backbone of the proposed frameworks in balancing the data distribution, while ensuring high-quality samples. These variants are discussed in the following subsection.

C. SMOTE VARIANTS

Numerous approaches have been proposed that combine SMOTE with undersampling, oversampling, ensemble, and algorithm-level techniques. These include widely adopted methods such as SMOTE-Tomek Links (SMOTE-TL), SMOTE with Rough Set Theory (SMOTE+RSB), SMOTE-WENN, and Safe-Level SMOTE [65], [74], [75], [76], as well as more recent variants like WSMOTER, OM-SMOTE, and SMOTE-kTLNN [77], [78], [79]. He et al. [78] proposed OM-SMOTE that offers better generalisation, while minimising intra-class imbalance on 32 datasets. The technique reduces class overlap by avoiding synthetic

data generation in overlapping regions. Kwat et al. [80] applied a hybrid technique that combines SMOTE-TL, VAE and conditional diffusion with deep learning models for augmenting ECG data. By incorporating MobileNetV2 and transformer models, they significantly improve classification performance, particularly for minority classes, making the system more effective for arrhythmia detection. Chitra et al. [81] proposed a two-phase framework to address imbalanced multi-label data. While Borderline MLSMOTE generates synthetic samples for minority classes in regions of high class overlap, an adaptive weighted L21-norm regularised logistic regression is then applied to enhance the prediction performance on the balanced dataset. An oversampling technique HHACO-FSOTe which generates synthetic minority samples using feature similarity was proposed by Sreeja et al. [82]. The technique considers all minority instances rather than choosing few neighbours for interpolation and is effective for low, high-dimensional and noisy datasets. Sun et al. [79] tackles noise associated with SMOTE by applying a two-layer nearest neighbour classifier (SMOTE-kTLNN) to an iterative-partition filter. After oversampling the minority class, kTLNN is used to filter noisy data, ensuring more accurate classification. Guo et al. [83] proposed an adaptive support vector-based B-SMOTE method that maps the dataset into a kernel-induced feature space and uses an SVM to identify support vectors. Minority class oversampling is performed by identifying neighbors based on kernel-defined distances. The SVM decision function is then constructed using kernel similarity values between samples, avoiding the need to explicitly generate synthetic data points in the high-dimensional space. A noted drawback of their method, however, is the relatively high execution time when dealing with large datasets. An ensemble of SMOTE, RF and SHAP was applied by Huang et al. [84] to identify and predict

decision-making mechanisms and susceptibility of slope geohazards. Chachoui et al. [85] leveraged on the strengths of four oversampling techniques, namely, SMOTE, BSMOTE, SMOTE-ENN and ADASYN to enrich the training data, while applying balanced bagging to generate multiple sub-models from the balanced subsets of the oversampled data. Being a combination of various SMOTE variants, the approach also incorporates their respective potential limitations and biases. Wen et al. [86] combined graph-based SMOTE with GNN and transformer-based feature extraction to address class imbalance in fraud detection. The method enhances node representations by integrating both structural and attribute features and uses synthetic node generation to improve minority class detection. Several works [87], [88], [89], [90] have also highlighted some clustering based oversampling method to tackle noise issues associated with SMOTE. Chen et al. [89] proposed an adaptive sampling with clustering and filtering noisy oversampling (ASCFNO) technique that combines clustering and noise filtering to address between-class and within-class imbalance issues. By utilising a density peak clustering with improved noise filter algorithm, the method effectively identifies and removes noisy and overlapping data before oversampling, leading to improved performance in imbalanced classification tasks. In addressing multi-class imbalance, Abdi et al. [53] introduced MDOBoost, which integrates Mahalanobis distance-based oversampling (MDO) with AdaBoost to improve classification performance. The MDO component generates synthetic samples that preserve the Mahalanobis distance (a statistical measure that accounts for class variance and feature correlation) from the class mean. This approach helps retain the underlying structure of each class, improving the model's ability to learn from imbalanced data. Han et al. [51] developed a global-local-based oversampling (GLOS) method to address multi-class imbalance by using three strategies: instance selection, partitioning, and oversampling. The method generates synthetic instances for isolated minority classes by accounting for the covariance and mean of all majority classes, ensuring equal distance to multiple majority class means. It also adapts instance generation through linear interpolation, shifting decision boundaries towards majority classes based on local density, and applies radial-based random oversampling within sub-clusters to expand the local space of minority classes. In the field of anomaly detection, Manokaran et al. [62] proposed a hybrid balancing technique that combines SMOTE with PPFCM to the noise of SMOTE. In their work, they clustered pre-processed datasets using PPFCM and hybridise the optimum cluster samples with SMOTE to create a balanced dataset. They considered other anomaly detection studies which addressed class imbalance. Ye et al. [91] proposed an oversampling framework that uses Laplacian eigenmaps—a nonlinear dimensionality reduction technique that preserves the local structure of data—to project the dataset into a lower-dimensional space where class separation is enhanced. This helps reduce the noise often introduced by SMOTE.

Building on this, a hybrid system combining SVM and Apache Spark was introduced [92], in which BSMOTE and Tomek links are applied during preprocessing to address class imbalance, followed by classification using SVM within Spark's distributed computing framework. In terms of clustering-based techniques, Zhao et al. [93] proposed a clustering-based oversampling algorithm (COM) to handle multi-class imbalance learning. In order to avoid the loss of important information, COM clusters the minority class based on the structural characteristics of the instances, among which rare instances and outliers are carefully portrayed through assigning a sampling weight, to each of the clusters. A summary of the highlighted emerging SMOTE variants for tackling imbalance is presented in TABLE 3. The corresponding abbreviations used for models, techniques, and metrics are defined in the Supplementary Material (Table S1).

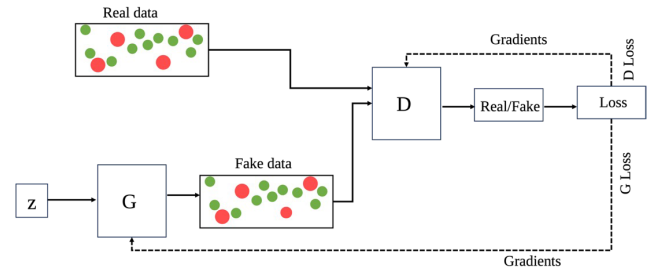


FIGURE 3. An illustration of GAN for data-augmentation.

D. GAN VARIANTS

GAN models comprises two neural networks: the generator (G) and the discriminator (D), in which G produces synthetics data based on a random input noise vector z , such that the generated data mimics the real data sample distribution [70], [105]. Evaluation of input data is achieved by D , which also classifies the data as either real (*i.e.* coming from the true data distribution) or fake (produced by the G). The training process is adversarial, whereby G , focuses on enhancing the quality of data that can fool D . Accordingly, D continually sharpens its ability to correctly distinguish between real and synthetic samples, leading to a dynamic in which both networks iteratively improves through a process of adversarial learning. The adversarial optimisation problem can be formulated as a min max game between G and D , with the objective function governing the process expressed as:

$$\min_G \max_D V(D, G) = \mathbb{E}_{x \sim P_{data}(x)} [\log D(x)] + \mathbb{E}_{z \sim P_Z(z)} [\log(1 - D(G(z)))] \quad (2)$$

where, $\mathbf{x}$ represents real data samples, $G(z)$ is the data generated by G from random noise z , and $P_{data}(x)$ and $P_Z(z)$ denote real data and noise distribution inputs respectively. The illustration in Fig. 3 visually represents the adversarial relationship between the generator (G) and discriminator (D) in GANs. It highlights how G produces synthetic data resembling real samples, while D evaluates and classifies data as real or fake. Through this iterative process, both

TABLE 3. Summary of notable SMOTE variants for imbalance learning.

Ref	Technique	Model	Metric	Dataset	Compared Methods	Category
Engine fault						
[94]	SMOTE/k-means	LGBM, XGB, SVM	F1, AUC, G-mean	Bearing, two moons, five_cluster, D31	DANN, MACGAN, Imba-DA, K-means-SMOTE, DPC-SMOTE	Multi.
Education						
[85]	EFD-based SMOTE	SVM, RF, DT, kNN, NB	Acc., prec., rec., F1, AUC	Student assessment	SMOTE, B-SMOTE, SMOTE-ENN, ADASYN	Multi.
Finance/fraud						
[95]	SMOTE + RUS	RF	Sens., spec., Acc., G-mean, F1	CSMAR	Bagging-DT, Ada-DT, DT, BP, Logit, Z-Score	Bin.
[86]	GTSMOTE	GNN, transformer	F1, AUC, G-mean	YelpChi, Amazon	GCN, GAT, GraphSAGE, CARE-GNN, PC-GNN, AO-GNN, H2-FDetector	Bin.
Generic						
[51]	GLOS	C4.5	DID, G-mean, prec., rec., CBA	KEEL, UCI	SMOTE, B-SMOTE, ADASYN, MDO, MC-BRO	Multi.
[52]	AMDO	DT	MAUC, G-mean, rec., prec.	KEEL, UCI	SMOTE, Ada.M1, SMB, OAO, OAA	Multi.
[78]	OM-SMOTE	SVM, DT, LR, NB	AUC, G-mean, F1, Accuracy	KEEL	SMOTE, B-SMOTE, ADASYN, Random-SMOTE, SMOTE-IPF, AMSCO, k-means SMOTE, G-SMOTE, SyMProD, LoRAS, AE-ELM-SynMin	Bin.
[81]	Borderline MLSMOTE	Elastic net	Hamming Loss, Subset Accuracy, Average Precision, Ranking Loss, One-error, Coverage	MULAN	IBLR, ECC, BR, RAKEL, ML-kNN, CLR, MLSR, MDDM, MORP	Multi.
[82]	HHACO-FSOTe	AdaBoost, LR, DT, KNN, GB, RF	AUC, G-mean	KEEL, UCI	B-SMOTE, ADASYN, AB-SMOTE, k-means SMOTE, RSMOTE, CS-SVM, SVDD, RUSBoost	Bin.
[79]	SMOTE-kTLNN	KNN, AdaBoost, SVM, C4.5	Rec., AUC, F1, G-mean	UCI, KEEL	SL-SMOTE, k-means SMOTE, SMOTE-RSB, SMOTE-IPF, SyMProD, SMOTE-ENN	Bin.
[83]	SV-B-SMOTE-SVM	SVM	G-mean, AUC, F1	UCI, KEEL	IFTSVM-ID, -ART- IFTSVM, DLSSVM-CIL, 1-NN, AdaBoost, EasyEnsemble, SVM-RUS, SVM-OSS, EFSVM, SMOTE, K-SMOTE-SVM, SMOTE-SVM	Bin.
Geology						
[84]	SMOTE-RF-SHAP	RF	Prec., rec., acc., F1, AUC	Geospatial	-	Multi.
Intrusion/ IoT/Social networks/Software						
[69]	SMOTE	RF, DT, KNN, LR, SVM, NB	Acc., rec., prec., F1, MCC	WUSTL-IIOT-2021	SMOTE, CSL, RUS, ROS, B&P	Bin./ Multi.
[96]	MWMOTE	GBM, XGBoost, LightGBM, Cat-Boost	MCC, AUC, F1	ORNL	-	Bin.
[97]	k-means SMOTE, enhanced RF	KNN, SVM, RF	Acc., prec., rec., F1	NSL-KDD	RepTree, I-ELM, A-PCA, RF, Ensemble (SVM, IBK, MLP)	Bin./ Multi.
[98]	GIWRF-SMOTE	LR, NB, SVM, DT, RF, KNN	Acc., prec., sens., F1, ROC, MCC	NSL-KDD	RUS, Near miss, ROS, ADASYN, SMOTE	Bin.
[99]	RUS+SVM SMOTE	NB, DT, LR, RF, GBM, KNN, SVM, MLP	Acc., F1, AUC, TPR, FPR	NSL-KDD, KDDCup99	OVS, SMOTENN, SMOTE, SMOTE-TL, BSMOTE, ROS, ADASYN, SVM, UDS AllKNN, IHT, RUS, ENN, C-Centroid, NearMiss, Condensed nearest neighbour	Multi.
[100]	GraphSMOTE	GNN	Acc., AUC, F1	Cora, BlogCatalog, Twitter	OVS, SMOTE, Embed SMOTE, Graph GSMOTE, Reweight,	Multi
[101]	SMOTE +stacking	DT, RF, NB, JRIP	Acc., Spearman's score, MSE	Qualitas corpus	SMOTE, stack	Multi.

TABLE 3. (Continued.) Summary of notable SMOTE variants for imbalance learning.

[62]	PPFCM- SMOTE	DT, NB, KNN, SVM, RF	Acc., prec, rec, F-score	IoT network dataset (DS2OS), UCI	K-Means-SMOTE, FCM-SMOTE, DBSCAN- SMOTE, and LOF-SMOTE	Bin.
Medical						
[102]	KNSMOTE	DT, Bagging, RF, AdaBoost	Sens., spec.	UCI	SMOTE, GSMOTE, ADASYN-SMOTE,ANS-SMOTE,MDO-SMOTE, SOMO, SOI_CJ, MWMOTE, k-means SMOTE	Bin.
[61]	SMOTE + LOF	DT, NB, SVM	Acc., prec., rec., F1, AUC	Kaggle, KEEL	SMOTE	Bin.
[103]	Incremental SMOTE	DT, KNN, LR, NB	Acc., prec., rec., F1, AUC	PM2.5	SMOTE, G-SMOTE, Gamma-SMOTE, SDD-SMOTE, TVAE synthesiser	Bin.
[80]	SMOTE-TL, VAE, CD	MobileNetV2, Transformer, CNN	Prec., Sens., F1, Acc.	MIT-BIH	Faster R-CNN, CNN-BLSTM, ArthyNet, SVM-RBF, CNN-LSTM	Multi.
Structures						
[104]	SMOTE	LR, DT, RF, SVM, KNN, GNB	Acc., prec., rec., F1	SSR - RS2	LIME	Multi.

networks improve, ultimately enabling the generation of realistic synthetic data for tasks such as data augmentation.

Several variants such as WGAN-GP, LSGAN, HingeGAN have been proposed for addressing class imbalance [106], [107], [108], [109], while enhancing sample generation, improving model performance and optimisation in various settings. Ahmed et al. [110] applied association rule in optimising majority class samples, while generating minority class data using a modified TGAN. Liu et al. [111] integrated wavelet transforms and capsule networks in enhancing fault diagnosis in imbalanced bearing datasets. Their method preserves essential signal features, while generating synthetic samples. Pu et al. [112] proposed a technique called Generative Adversarial One-off Diagnosis (GAOSD) to reduce the dependence on large datasets in fault diagnosis. They used a Bidirectional GAN for both feature extraction and data generation on healthy samples, as standard GANs are not well-suited for handling time-series data. Wang et al. [113] introduced a dual-module multi-head spatiotemporal joint network for addressing class imbalance in wind turbine fault detection. The work combines auxiliary classifier GAN for generating high-quality fault samples and integrates CNN and LTS for spatiotemporal representation. Furthermore, Gu et al. [114] introduced CSWGAN which improves the generative process and results by integrating cosine similarity-based penalty terms to regulate the optimisation objectives. Das [115] proposed the combination of undersampling and oversampling based on rule mining and GAN *i.e.*, majority class duplicates are eliminated using rule mining while minority class are resampled using TableGAN. Thus, addressing the risk of overfitting and data loss characterised by the former methods. The superiority of their method is compared with emerging data-augmentation techniques [6], [116], [117]. Lin et al. [8] developed a novel multiple correlation analysis (MCA) which integrates several coefficients for redundancy and sensitivity

analysis for selecting key features in high-dimensional, highly-imbalanced data. They combined MCA with a developed GAN variant (1DWGANGP) for data-augmentation in tackling imbalance and predicting the interval ranges of vibration in an aero-engine assembly dataset. Ding et al. [118] proposed LEGAN to address issues of intraclass model collapse associated with GAN during sample generation. A summary of GAN variants for highlighted is presented in TABLE 4.

E. COMPARISON OF SMOTE AND GAN VARIANTS FOR DATA-AUGMENTATION

While both SMOTE and GAN variants aim to enrich datasets and improve model performance, they differ significantly in their methodologies, strengths, and limitations. TABLE 5 provides a detailed comparison between SMOTE and GAN variants for data augmentation, focusing on their methods, applicability, strengths, and limitations across different criteria.

GAN-based methods have shown advantages over SMOTE-based variants in modelling complex or multi-modal class distributions, particularly in high-dimensional or unstructured datasets [9], [72], [105], [111]. They often generate more diverse and realistic synthetic samples that can help improve classifier generalisation [71], [111], [114], [118]. However, their performance is sensitive to training configurations and typically requires more data and computational resources, along with a risk of mode collapse if not properly tuned [70], [71], [105]. In contrast, SMOTE and its variants remain effective and efficient for lower-dimensional, tabular data, offering fast preprocessing and ease of integration into existing pipelines [31], [59], [60], [63]. These methods are especially useful when training speed, interpretability, or limited computational budgets are primary concerns. Overall, the choice between SMOTE-based and GAN-based augmentation should depend

TABLE 4. Summary of notable GAN variants for imbalance learning.

Ref	Technique	Model	Metric	Dataset	Compared Methods	Category
Engine fault						
[119]	BCTGAN	CNN	Acc., AUC	CWRU, GFDS2, self-mined	SMOTE, VAE, GAN, LinearGAN, 2dDCGAN, WGAN-GP, LSGAN, HingeGAN	Multi.
[8]	MCA + 1DWGANGP	KNN, MLP, DT, SVM	Acc.	UCI, Aero-engine assembly, Real estate	SMOTE, MCA+ WGAN, LSGAN, CGAN, DCGAN	Multi.
[120]	DAGAN	ResNet	Bal. Acc., F1, AUC	Bearing, gearbox	RUS, CC, ROS, SMOTE, MC-SMOTE, ADASYN, DCGAN, CD-CGAN, DDPM, DCGAN-TL	Bin.
[111]	WCGAN	SVM, SAE, BNCNN	F1	Bearing	UDS, OVS, VAE, WGAN-GP	Multi.
[114]	CSWGAN-GP	CNN	Acc., G-mean, F1, rec., PRC, ROC	Bearing	SAGAN, ACGAN, WGAN-GP, DFPCNN, B-SMOTE, SMOTE	Multi.
[112]	GAOSD	RF	Prec., rec., F1	self-mined (WT901C-485 attitude sensor)	OSD-WFE, OSD-SAE, OSD-DCAE	Multi.
[113]	DMSJN-SACGA	CNN-LSTM	Acc., Macro-F1, G-mean-F1, prec., recall	3 GW WT	ACGAN, CGAN, SGAN	Multi.
Generic						
[110]	RAMGANH	SVM	AUC, F1	KEEL	EUSCHC, SMOTE, ADASYN, S-SMOTE, SMOTE-TL, SMOTE-RSB	Bin.
[115]	Rule mining + TGAN	KNN, SVM, RF	Acc., sens., F1	UCI, KEEL	Model fusion (ANN, Ada, RF, LR), OBU, EUSC, Laplacian eigenmap + SMOTE	Bin.
Intrusion/IoT/Social networks/Software						
[121]	CTGAN	DT, SVM, NB, RF, FNN, LSTM, CNN	Acc., prec., rec., F1, MCC	NSL-KDD, UNSW-NB15	PB-DID, CNN-BiLSTM	Multi.
[122]	KNN+ TACGAN	MLP, KNN	Acc., F1, AUC, rec.	KDDCUP99, UNSW-NB15, CICIDS2017	ROV, DAN, SMOTE, SMOTE-ENN, ADASYN, CGAN, WGAN, ACGAN, MAGENTO, IGAN-IDS	Bin./ multi.
[123]	TDCGAN	MLP	Acc., prec., rec., F1	UGR'16	SMOTE, ROS, SMOTEENN, BSMOTE, SVM-SMOTE, SMOTE-TL, SMOTE_NC, CGAN, CTGAN	Multi.
Medical						
[118]	LEGAN	Resnet	Prec., rec., F1, acc.	BloodMNIST, OrgancMNIST, PathMNIST	LOF, GAN, GAN+LOF, GAN+EL, GAN+AL, DCGAN, WGAN-GP, SNGAN, StyleGAN2, SMOTE	Multi.

on the nature of the dataset and the specific goals of the application.

V. EVALUATION METRICS FOR IMBALANCE PROBLEMS

Aside conventional metrics like Accuracy (Acc.), precision (prec.), specificity (spec.), recall (rec.), F1-score/F-measure. Metrics adopted in evaluating model performance where class-imbalance exist include AUC, Mean Area Under the Curve, Discreteness-based Imbalanced Degree, Geometric mean, Extended Geometric mean, Matthews Correlation Coefficient and Imbalanced Multi-class Classification Performance.

A. AREA UNDER THE RECEIVER OPERATOR CHARACTERISTICS CURVE

The Area Under the Receiver Operating Characteristic Curve (AUC) is derived from the confusion matrix but differs from other metrics in that it is independent of any specific decision

threshold. Instead, it evaluates model performance across all possible thresholds by plotting the True Positive Rate (TPR) against the False Positive Rate (FPR) [18], [125]. Since both TPR and FPR are computed from the confusion matrix, AUC still reflects the model's classification behaviour. A higher AUC indicates a better balance between TPR and FPR, with a value of 1 representing a perfect classifier. The ROC curve visualises how these rates change with different classification thresholds. These rates are defined as:

$$\text{TPR} = \text{Sensitivity} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (3)$$

$$\text{FPR} = 1 - \text{Specificity} = \frac{\text{FP}}{\text{FP} + \text{TN}} \quad (4)$$

where TP (True Positive) and TN (True Negative) represents the cases where the model correctly predicts the positive and negative classes respectively; while FP (False Positive) and FN (False Negative) are cases where the model incorrectly predicts the positive and negative instances.

TABLE 5. A detailed comparison of SMOTE and GAN variants for data augmentation.

Criteria	SMOTE variants	GAN variants
Computation cost	Comparatively resource efficient	Slower, computationally resource-intensive
Data augmentation strategy	Linear interpolation between nearest neighbours	Adversarial training using a generator and discriminator
Ease of use	Simple, quick, and easy to implement	Harder to implement, requires extensive tuning and training
Flexibility	Limited to tabular data	Highly flexible, capable of handling structured, unstructured, and image data
Risk of overfitting	Higher in high-dimensional data	Lower with careful configuration and training with risk of mode collapse in specific cases
Robustness to noise	Susceptible to noise in the data	Can model noise effectively but requires careful training to avoid artefacts
Synthetic data quality	Limited diversity, less realistic	High diversity, more realistic synthetic data
Use case	Suitable for simple, low-dimensional, tabular datasets	Suitable for complex, high-dimensional, unstructured datasets

B. MEAN AREA UNDER THE CURVE

Mean Area Under the Curve (MAUC) or M-measure [125] is an extension of the AUC metric designed for multi-class classification. It generalises to multi-class problems by averaging the AUC across all class pairs, offering a single metric that captures the classifier's ability to distinguish between multiple classes and expressed as:

$$MAUC = \frac{2}{n(n-1)} \sum_{i=1}^{n-1} \sum_{j=i+1}^n AUC(i, j) \quad (5)$$

where n and $AUC(i, j)$ are the number of classes and the AUC score for the pair of classes i and j , respectively.

C. DISCRETENESS-BASED IMBALANCED DEGREE

Han et al. [51] introduced a discreteness-based imbalanced degree (DID) metric designed to clearly differentiate between majority and minority classes. This metric also dynamically adjusts the number of synthetic instances generated for each minority class instance. By doing so, it aims to avoid the production of excessive redundant instances that could result in overfitting, while simultaneously improving the quality of the synthetic instances to the greatest extent possible. It is expressed as:

$$DID = \sum_{j=1}^m \theta_j \frac{C_j}{N} \quad (6)$$

where C_j is the number of instances in class j , N is the total number of instances, and θ_j reflects how spread out the samples are within class j . A lower θ_j indicates that the class is more sparsely distributed, making it more difficult for the model to learn from. DID uses this information to adapt the number of synthetic samples generated for each class, helping to reduce overfitting while improving sample quality.

D. GEOMETRIC MEAN

The G-mean, or geometric mean, is commonly employed in binary classification tasks to ensure balanced performance between the positive and negative classes. It is calculated by taking the square root of the product of the true positive rate (sensitivity) and the true negative rate (specificity).

$$G\text{-mean} = \sqrt{TPR \cdot TNR} \quad (7)$$

with TNR defined as:

$$TNR = \frac{TN}{TN + FP} \quad (8)$$

G-mean is effective in ensuring that a model performs well on both classes, particularly in imbalanced datasets where accuracy alone might be misleading. A higher G-mean, closer to 1, indicates that the model is performing well on both classes, striking a balance between identifying minority class instances and correctly classifying majority class instances.

E. EXTENDED G-MEAN

The extended G-mean is a modified version of the G-mean tailored for multi-class classification problems. Unlike the standard G-mean, which is designed for binary classification, the extended G-mean broadens this concept to accommodate multiple classes. This metric generally involves calculating the G-mean for each class individually, considering the model's performance across all classes, and then combining these results. In a multi-class setting, the extended G-mean can be computed by taking the geometric mean of the sensitivity for each class and is given as:

$$\text{Extended G-mean} = \left(\prod_{i=1}^n \text{Sensitivity}_i \right)^{\frac{1}{n}} \quad (9)$$

where n is the number of classes and Sensitivity_i is the sensitivity of class i .

F. MATTHEWS CORRELATION COEFFICIENT

Matthews Correlation Coefficient (MCC) is a robust evaluation metric in binary classification, suited for scenarios involving class imbalance. The MCC takes into account all four quadrants of a confusion matrix: true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN), to produce a single coefficient that reflects the balance between the positive and negative classes. It is approximated as:

$$MCC = \frac{(TP \cdot TN) - (FP \cdot FN)}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}} \quad (10)$$

Since MCC accounts for both correct and incorrect predictions across all classes, MCC delivers a balanced

evaluation even when one class significantly outweighs the other. This makes it a more dependable alternative to commonly used metrics like accuracy or F1 score, offering a comprehensive reflection of a model's true predictive capability in imbalanced learning environments.

G. IMBALANCED MULTI-CLASS CLASSIFICATION PERFORMANCE

Imbalanced Multi-class Classification Performance (IMCP) is an open-source Python package by Aguilar-Ruiz et al. [126] designed to evaluate and visualise model performance on multi-class imbalanced datasets. Unlike AUC, IMCP directly handles multi-class datasets without requiring extensions like one-vs-all or one-vs-one strategies. It uses prediction probabilities instead of confusion matrices, emphasising minority classes to provide a more accurate assessment of classifier performance. The IMCP curve integrates class distribution and Hellinger distance, with contributions inversely proportional to class size, offering clearer insights into how classifiers manage imbalanced data. Further analysis is needed to validate its effectiveness across diverse datasets and imbalance severities.

VI. DISCUSSION AND FUTURE STUDIES

The recently developed SMOTE variants demonstrate the versatility and adaptability of the original technique in addressing imbalanced datasets across various domains. By avoiding regions where classes overlap and by pairing SMOTE with more advanced models, researchers are reducing intra-class imbalance and filtering out noise more effectively. The application of ensemble methods further highlights the importance of combining variants in mitigating the shortcomings of SMOTE. Despite their effectiveness, some of these SMOTE variants still pose challenges in terms of computational complexity and resource expense, especially with high-dimensional and large datasets. Interestingly, recent findings in radiomics [127] reveal that applying the common SMOTE variants did not consistently improve predictive performance across datasets. Although slight improvements (e.g., +0.015 in AUC) were observed in specific cases, the overall impact was minimal. This calls attention to the need for context-specific evaluation of data augmentation techniques.

Given the increasing use of emerging SMOTE and GAN variants, it is crucial to investigate whether such trends hold true on them, especially in complex and data-sensitive domains. Future studies could therefore compare SMOTE-based resampling techniques with GAN-based data generation methods to determine whether these emerging variants offer measurable and consistent benefits, or if their impact similarly varies across datasets. Although, computationally expensive, the employment of GAN variants with careful training offers promising capability in handling multi-modal unstructured and structured data, when compared to its SMOTE counterparts. However, challenges such as computational complexity, model collapse,

non-convergence on multi-class large-scale high-dimensional datasets remain.

The aspect of mode collapse [128], where the generator ends up producing repetitive outputs and fails to represent the full diversity of the data, especially in minority classes is an area requiring significant attention. This not only affects how reliable the model is but also raises concerns about how well these techniques scale. The challenge becomes more pronounced when GANs are applied to other data types, such as tabular, time-series, or text. In these contexts, limited sample diversity can amplify domain-specific biases and obscure critical class distinctions. It is therefore important to focus on designing GAN variants that can preserve diversity while adapting to different data domains.

To advance the field of class imbalance handling in ML, key areas should be the focus of future research:

- 1) *Scalability and large-scale datasets*: Enhancing the scalability and generalisation capabilities of existing approaches is crucial for managing large-scale datasets effectively. Future work should aim to reduce computational complexity, making them more practical in real-world use. Expanding the techniques to address multi-class issues could significantly broaden their applicability. By overcoming these limitations, the proposed methods can become more adaptable and efficient in handling class imbalances across a variety of datasets.
- 2) *Impact of data quality*: The effectiveness of class imbalance solutions is often influenced by the quality of the original dataset. High levels of noise or errors can diminish the optimisation and synthesis phases of algorithms. Future studies should explore strategies to mitigate the negative effects of poor-quality data, such as implementing robust preprocessing steps or developing algorithms resilient to noise and errors.
- 3) *Integration with standard SMOTE variants*: While many new methods claim superiority over standard techniques like SMOTE, it is essential to incorporate these approaches within a unified framework for direct comparison. Integrating each technique into a pipeline similar to that of SMOTE and conducting statistical tests across different application domains can provide deeper insights into their effectiveness. This comparative analysis would help identify the most robust methods, specificity towards domain requirements and potentially reveal synergistic effects when combining techniques.
- 4) *Training Stability and Diversity*: Developing GAN variants that effectively address challenges like mode collapse, non-convergence, and scalability in multi-class, high-dimensional datasets is still a prevailing issue. Emphasis should be placed on preserving data diversity and adapting generative models to complex domains such as tabular, time-series, and text data, where maintaining class distinction is critical. Comparative evaluations with conventional resampling

methods will be key to validating their effectiveness and domain applicability.

- 5) *Comprehensive evaluation of IMCP performance*: A thorough analysis is necessary to build confidence in the performance of the IMCP curve across diverse datasets and varying degrees of class imbalance. Future research should involve extensive testing to assess the robustness and reliability of IMCP in different scenarios. Such an evaluation would not only validate its effectiveness but also help in understanding its limitations and areas for improvement.
- 6) *Enhance model validation performance in rarity conditions*: A robust validation framework and alternate metrics that specifically account for randomness and chance in minority sample rarity conditions could be developed. The data-augmentation techniques could also be tailored to rare class scenarios that reduce the effect of randomness. These advancements will improve the effectiveness of ML models in critical real-world applications where imbalance exists. By addressing these research areas, the field can develop more robust, scalable, and adaptable solutions for class imbalance, ultimately enhancing the performance of ML models in numerous applications.

VII. CONCLUSION

Class imbalance often biases models towards majority classes, resulting in poor recognition of patterns in minority classes. This review explores emerging data-augmentation techniques, specifically SMOTE and GAN, which have gained attention for addressing imbalance and overcoming limitations of common approaches. Reliable evaluation metrics for varying imbalance scenarios are also discussed. While significant progress has been made, future research should prioritise improving scalability, validation under rare sample conditions, and the integration of hybrid techniques and noise-resilient algorithms to address real-world challenges. This review offers new insights and identifies directions for informing future solutions to class imbalance in ML tasks.

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